

No Cubic Annihilators for Canonical Coordinates of Permutations of $\mathbb{F}_2^4$: A Certified Finite Exclusion

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Abstract

Every map $F : \mathbb{F}_2^4 \rightarrow \mathbb{F}_2^4$ has unique multilinear coordinate representatives $f_1, \dots, f_4$. We prove, by an exact computer-assisted argument, that if F is a permutation and $H \in \mathbb{F}_2[U_1, \dots, U_4]$ has total degree at most three, then the formal identity $H(f_1, \dots, f_4) = 0$ in the ordinary polynomial ring implies $H = 0$. The proof identifies the 20-dimensional space of cubic polynomials vanishing on the Boolean output grid, partitions its $2^{20} - 1$ nonzero elements into 48 affine orbits, removes seven reducible orbits by a degree-two argument, and excludes the remaining 41 representatives with independently checked DRAT certificates. Consequently, any annihilator for such a permutation has degree at least four. This restricted result does not decide whether the four canonical coordinates are algebraically independent.

1 Scope and context

Put

$$R = \mathbb{F}_2[x_1, x_2, x_3, x_4], \quad S = \mathbb{F}_2[U_1, U_2, U_3, U_4].$$

For a set map $F : \mathbb{F}_2^4 \rightarrow \mathbb{F}_2^4$, write $f = (f_1, \dots, f_4)$ for the unique coordinatewise multilinear lift of F . An *annihilator* is a nonzero $H \in S$ for which $H(f) = 0$ as an identity in R ; no Boolean reduction in the source variables is allowed.

Barańczuk and Ślusarski study algebraic independence of polynomial representatives of finite-field maps. Among other results, they prove algebraic independence for permutation representatives in dimensions one and two and verify by exact exhaustive computation that the canonical 2-reduced representatives of all permutations of $\mathbb{F}_2^3$ have algebraically independent coordinates [1]. Their general finite-grid argument excludes annihilators of total degree strictly less than the field size. In the binary permutation setting, the elementary domain argument below strengthens this through degree two; our certified result advances it through degree three at the next dimension. Our comparison is limited to the cited source and is not a claim of global priority.

Theorem 1. *Let $F : \mathbb{F}_2^4 \rightarrow \mathbb{F}_2^4$ be a permutation and let $f_1, \dots, f_4 \in R$ be its canonical multilinear coordinate representatives. If $H \in S$ has total degree at most three and*

$$H(f_1, f_2, f_3, f_4) = 0$$

in R , then $H = 0$.

The theorem gives only a lower bound of four on the degree of a possible annihilator. In particular, it neither excludes relations of degree at least four nor resolves the motivating algebraic-independence question.

2 The grid kernel through degree three

We first record the non-computational degree-two obstruction in the specialization needed here. Write $q_i = U_i^2 + U_i$.

Lemma 2 (degree-two obstruction). *If F is a permutation of $\mathbb{F}_2^4$, $\deg H \leq 2$, and $H(f) = 0$, then $H = 0$.*

Proof. The identity implies that H vanishes at every point of $\mathbb{F}_2^4$, since F is surjective. The vanishing ideal of the Boolean grid is (q_1, q_2, q_3, q_4) . Its degree-at-most-two part consists precisely of the constant linear combinations of the q_i , so $H = \sum_i c_i q_i$ for some $c_i \in \mathbb{F}_2$. After substitution, with $g = \sum_i c_i f_i$, we obtain

$$0 = g^2 + g = g(g + 1).$$

Because R is an integral domain, g is the constant zero or the constant one. If $(c_1, \dots, c_4) \neq 0$, however, the function $x \mapsto g(x) = \sum_i c_i F_i(x)$ takes both values on the grid because F is surjective. This contradiction forces every c_i to be zero. $\square$

Let $S_{\leq 3}$ denote the polynomials of total degree at most three and let

$$\text{ev} : S_{\leq 3} \longrightarrow \{\mathbb{F}_2^4 \rightarrow \mathbb{F}_2\}$$

be Boolean-grid evaluation.

Lemma 3 (cubic grid kernel). *The kernel of ev has dimension 20 and has the basis*

$$q_i, \quad q_i U_j \quad (1 \leq i, j \leq 4).$$

Consequently every member of this kernel has a unique expression

$$H = \sum_{i=1}^4 q_i \left(c_{i,0} + \sum_{j=1}^4 c_{i,j} U_j \right), \quad c_{i,j} \in \mathbb{F}_2. \quad (1)$$

Proof. There are $\binom{7}{4} = 35$ monomials in four variables of total degree at most three. On the Boolean grid, a monomial is identified with the squarefree monomial having the same support. The 15 squarefree monomials whose supports have size at most three are linearly independent as Boolean functions. Hence the evaluation map has rank 15 and its kernel has dimension 20.

The displayed 20 polynomials vanish on the grid. Their independence can be checked in disjoint support blocks. For singleton support $\{i\}$, the two elements $q_i = U_i^2 + U_i$ and $q_i U_i = U_i^3 + U_i^2$ are independent. For pair support $\{i, j\}$, the two elements $q_i U_j$ and $q_j U_i$ are independent. The singleton blocks contribute eight elements and the six pair blocks contribute twelve. They therefore form a basis. $\square$

Thus a prospective nonzero cubic annihilator is represented by a unique nonzero 20-bit coefficient mask in (1).

3 Affine reduction and reducible masks

The group $G = \text{AGL}(4, 2)$ acts on S by invertible affine substitution in the U variables. This action preserves total degree and the Boolean grid kernel. More importantly, it preserves the existence of a permutation witness: if $H(f) = 0$ and $T \in G$, then $g = T \circ f$ is the canonical multilinear lift of the permutation $T \circ F$, and $(H \circ T^{-1})(g) = 0$.

The exact bitset enumeration uses eight substitutions: the four coordinate translations, the three adjacent coordinate transpositions, and the transvection $U_1 \mapsto U_1 + U_2$. Adjacent transpositions conjugate the last map to the elementary coordinate additions; these, together with coordinate permutations, generate $\text{GL}(4, 2)$, and the four translations generate its translation subgroup. Breadth first search under these generators partitions all $2^{20} - 1 = 1,048,575$ nonzero masks into 48 orbits, with the computed orbit sizes summing to the full number of masks.

Reducibility is also decided by exact linear algebra. There are 30 nonconstant affine linear polynomials over $\mathbb{F}_2$: the linear part has 15 nonzero choices and the constant has two choices. For each orbit representative H and each such L , Gaussian elimination over $\mathbb{F}_2$ tests whether

$$H \in L S_{\leq 2}.$$

Every reducible nonzero polynomial of degree at most three has an affine linear factor, so this is exhaustive. Seven of the 48 orbits are reducible; together they contain 6,485 masks.

No mask in a reducible orbit can have a permutation witness. Indeed, if $H = AB$ is a nontrivial factorization and $H(f) = 0$, the integral-domain property of R gives $A(f) = 0$ or $B(f) = 0$. One factor has degree at most one and the other at most two, contradicting Lemma 2. It remains to exclude 41 irreducible representatives, whose orbits contain the other 1,042,090 masks.

4 Exact propositional encoding

Fix one of the remaining representatives H . For $p \in \mathbb{F}_2^4$ and $1 \leq i \leq 4$, introduce a Boolean variable $y_{p,i}$ for the i th bit of $F(p)$. The constraints

$$\bigwedge_{p \neq p'} \bigvee_{i=1}^4 (y_{p,i} \oplus y_{p',i}) \quad (2)$$

state that the 16 output words are pairwise different. Since there are exactly 16 possible words, this is equivalent to F being a permutation.

Identify a support $A \subseteq \{1, 2, 3, 4\}$ with its indicator point $\mathbf{1}_A \in \mathbb{F}_2^4$. The Boolean Möbius transform gives the exact ANF coefficient

$$a_{i,A} = \bigoplus_{B \subseteq A} y_{\mathbf{1}_B, i}, \quad f_i = \sum_{A \subseteq [4]} a_{i,A} x^A. \quad (3)$$

These equalities are imposed propositionally.

The program then expands

$$H(f) = \sum_{i=1}^4 (f_i^2 + f_i) \left(c_{i,0} + \sum_{j=1}^4 c_{i,j} f_j \right) \quad (4)$$

as an ordinary polynomial in the source variables. In characteristic two, $f_i^2 = \sum_A a_{i,A} x^{2A}$, and the exponent of each source variable in (4) lies in $\{0, 1, 2, 3\}$. Equal terms are cancelled modulo two,

and every surviving formal coefficient is constrained to zero. The constant ANF atom cancels from $f_i^2 + f_i$ because $a_{i,\emptyset}^2 + a_{i,\emptyset} = 0$ for a Boolean coefficient.

Proposition 4 (semantic equivalence). *The Boolean formula consisting of (2), (3), and all formal coefficient constraints from (4) is satisfiable if and only if there exists a permutation F whose canonical multilinear coordinates satisfy $H(f) = 0$ in R .*

Proof. From a satisfying assignment, (2) defines a permutation table. Formula (3) is the inverse Boolean evaluation transform, hence gives exactly the canonical multilinear coordinates of that table. The remaining constraints set every ordinary-polynomial coefficient of $H(f)$ to zero, so $H(f) = 0$ formally. Conversely, the table of any witnessing permutation, together with its ANF coefficients, satisfies each of these constraints. No implication uses Boolean reduction of the source monomials. $\square$

All atoms in this formula are Boolean. Conjunctions and parity constraints are converted to DIMACS by Tseitin definitions. Thus a DRAT refutation verifies unsatisfiability of the resulting propositional CNF; it is not being used to reason about opaque arithmetic atoms. This use of proof-producing SAT after encoding compression is also found in an unrelated finite combinatorial setting [3]; that citation is a methodological comparison only, not prior work on finite-field polynomial maps.

5 Certificates and conclusion

The all-different and ANF constraints without a relation were first checked satisfiable. For each of the 41 irreducible representatives, the full formula was translated to CNF. CaDiCaL 3.0.1 returned UNSATISFIABLE and emitted a DRAT proof, and `drat-trim` verified every proof. The role of DRAT checking is the standard one of independently replaying clausal unsatisfiability certificates [2].

Finite object	Exact count or range
Nonzero masks in the cubic grid kernel	1,048,575
AGL(4, 2) orbits	48
Reducible orbits (masks)	7 (6,485)
Irreducible orbits (masks)	41 (1,042,090)
CNF clauses per irreducible representative	3,154–12,894
Verified DRAT refutations	41 of 41
Independent formal-support comparisons	3,584

The separate support audit does not reuse the coefficient-term expansion: it constructs ANFs from permutation tables and performs direct $\mathbb{F}_2$ polynomial-set multiplication. It compared all 48 orbit representatives and 64 additional deterministic masks on 32 permutations each, for 3,584 exact matches. These comparisons are a diagnostic check on the implementation; the semantic justification is Proposition 4, together with inspection of the exact generator source. The certificate manifest records the instance sizes and the result of each proof check.

Proof of Theorem 1. Any degree-at-most-three annihilator vanishes on the output grid and hence, by Lemma 3, corresponds to one of the nonzero 20-bit masks. Affine invariance moves it to one of the 48 representatives. The seven reducible representatives are impossible by Lemma 2 and the integral-domain factor argument. For each of the other 41, Proposition 4 identifies a permutation witness with a satisfying assignment of the corresponding CNF, while its checked DRAT certificate proves that CNF unsatisfiable. All nonzero masks are therefore excluded. $\square$

6 Reproducibility and limitations

The proof archive contains the 41 DIMACS/DRAT pairs and a manifest. The following commands, run from the project directory, reproduce the exact finite searches and regenerate certificates in a temporary directory; they require Python with Z3 5.1.0, CaDiCaL 3.0.1, and `drat-trim` on the executable path.

```
PYTHONDONTWRITEBYTECODE=1 python3 evidence/verify_degree2.py
```

```
PYTHONDONTWRITEBYTECODE=1 python3 evidence/search_cubic.py \  
--timeout-ms 60000 --max-orbits 41 \  
--output /tmp/sm46-cubic-search.json
```

```
PYTHONDONTWRITEBYTECODE=1 python3 evidence/certify_cubic.py \  
--directory /tmp/sm46-cubic-certificates
```

```
PYTHONDONTWRITEBYTECODE=1 python3 evidence/audit_cubic_encoding.py \  
--output /tmp/sm46-cubic-encoding-audit.json
```

The DRAT files certify the generated CNFs. Their applicability to the stated polynomial theorem also depends on the explicit kernel, orbit, factor, ANF, and coefficient-expansion reductions given above and on the auditable source that implements them. Successful computation is not a claim about relations of degree four or higher. Accordingly, the algebraic-independence problem for canonical coordinates of permutations of $\mathbb{F}_2^4$ remains unresolved.

Computational and AI assistance. AI assistance was used in proof exploration, exact-program development and review, and preparation of this exposition. The mathematical claim rests on the stated reductions, the inspectable exact source code, the finite CNF instances, and replayable DRAT certificates rather than on model-generated assertions. The author takes responsibility for the statement and its scope.

References

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- [3] Zijian Zeng, Houde Liu, Kuru Ratnavelu, and Ong Seng Huat. One-sided parity compression for certified SAT attacks on the 33-point erdős-szekeres problem, 2026. Preprint available at SSRN, doi:10.2139/ssrn.7334758.